LESSON PLAN TEMPLATE

Name: **Cohort:**

Lesson Plan

Lesson Title: Multi-Day Lesson Plan: Circle Measurements – Grade 7 Ontario Mathematics	**Date:**
Grade Level: 7	**Subject/Strand: Math – Strand E (Spatial Sense / Measurement)**
Topic: Circles	**Length of Period: 50 minutes Number of Days: 4**

Lesson Plan Description -- What do I want the learners to know and/or be able to do? What are the big ideas/enduring understandings for the lesson/unit? Indicate using 1-2 sentences.

Learners will explore how circles can be measured and described using radius, diameter, circumference, and area. Across four days, students will use hands-on inquiry with ribbons to discover the constant ratio between circumference and diameter (π), learn formal circle terminology, apply formulas to calculate diameter, radius, and circumference, and derive the area formula by cutting circles into sectors and rearranging them into a rectangle/parallelogram shape. Students will understand that these measurements are interconnected and can be used to describe the size of any circle accurately.

STEP 1: CURRICULUM CONNECTIONS

Ontario Curriculum Overall Expectations (numbers from documents and details)	**Ontario Curriculum Specific Expectations** Numbers from documents and details -- a realistic number of expectations (1 or 2), connect to assessment.
E2. Measurement: Compare, estimate, and determine measurements in various contexts.	E2.3: Use the relationships between the radius, diameter, and circumference of a circle to explain the formula for finding the circumference and to solve related problems. E2.5: Show the relationships between the radius, diameter, and area of a circle, and use these relationships to explain the formula for measuring the area of a circle and to solve related problems.
Learning Goals **Discuss with learners: *What will I be learning today?*** Clearly identify what learners are expected to know and be able to do.	**Success Criteria** **Discuss with learners: *How will I demonstrate what I have learned?*** Teacher/learner identify 'look fors' in language that learners can readily understand.
We are learning … Discover the constant relationship between the circumference and diameter of a circle using hands-on measurement (Day 1). Identify and define key circle terminology: radius, diameter, circumference, centre, chord, and pi (Day 2).	I can … Measure the circumference and diameter of circular objects using a ribbon and describe the relationship I notice between them. Define and identify the radius, diameter, circumference, centre, and chord on any circle. Explain how the radius and diameter are connected (diameter = 2

Calculate the radius, diameter, and circumference of circles using formulas (Day 3). Derive and apply the formula for the area of a circle by cutting circles into sectors and rearranging them into a rectangle (Day 4).	× radius). Use the formulas $C = \pi d$ and $C = 2\pi r$ to calculate circumference. Calculate the radius when given the diameter (and vice versa). Explain why cutting a circle into sectors and rearranging them forms an approximate rectangle, and use this to explain $A = \pi r^2$. Solve problems involving the area of a circle.

STEP 2: ASSESSMENT

Indicate purpose of the assessment : [ ] ***FOR*** [ ] ***AS*** [ ] ***OF***

Day	Purpose	Type
Day 1	**Assessment FOR Learning** (Diagnostic/Formative)	Observing how students reason about measurement relationships before formal instruction
Day 2	**Assessment FOR Learning** (Formative)	Checking understanding of vocabulary through worksheet and discussion
Day 3	**Assessment AS Learning** (Formative)	Students self-assess calculation accuracy on worksheet; teacher circulates to provide feedback
Day 4	**Assessment OF Learning** (Summative/Formative)	Students demonstrate understanding of area derivation and solve area problems on worksheet

Indicate Achievement Chart categories being assessed
[X] ***Knowledge and Understanding*** **[X]*Thinking*** **[X]*Application*** **[X]** ***Communication***

Indicate Learning Skills/Work Habits:
[ ] responsibility, [ ] organization, **[X]** independent work, **[X]**collaboration,**[X]**initiative, [ ] self-regulation
(Incorporate one with Learning Goal and Success Criteria)

Indicate Assessment Mode:
Written, Oral, Performance
(Write, Say, Do)

Mode
Performance (Do): Hands-on ribbon measurement activity
Written (Write): Vocabulary worksheet, calculation worksheets, area worksheet
Oral (Say): Class discussion, partner talk, consolidation responses

Indicate Assessment Strategy : What will learners do to demonstrate their learning?

Strategy
Observation of student measurement techniques, data recording, and pattern recognition
Review of completed worksheets for accuracy and mathematical reasoning
Questioning during circulating and whole-class debrief

Indicate Assessment Tool: Instrument used to record results/ document learning

Tool
Anecdotal notes checklist
Worksheets (collected and reviewed)
Teacher observation notes

STEP 3: CONSIDERATIONS FOR PLANNING

Prior Learning: What prior experiences, knowledge and skills do the learners bring with them to this learning experience?

Students have completed an estimation unit involving mass, capacity, and linear measurement. They can use appropriate tools, estimate lengths, and justify unit choices. They have also worked with basic 2-D shapes but have not yet formally defined circle measurements. Students are familiar with area of rectangles and parallelograms from earlier grades.

IEP program implications: Accommodations, Modifications -- To be completed for Placements 2, 3 and 4

Differentiation -- How will I differentiate the instruction to ensure the inclusion of all learners? (Choose 1 or 2 areas in Year 1)

Content: All students work with the same core concepts; advanced students explore additional circles or more complex problem contexts. Struggling students receive pre-drawn diagrams with key points highlighted.

Process: Step-by-step visual instructions are available for hands-on tasks. Students who need more challenge compare measurements across multiple circles and describe patterns independently. Students needing support receive guided recording sheets with sentence starters.

Product: Some students demonstrate learning orally during discussion; others demonstrate through written worksheet responses. Students can draw, write, or explain verbally.

Environment: Flexible grouping — strategic pairing of students during partner work. Movement-based activities (Day 1 ribbon activity) support kinesthetic learners. Quiet independent work time is provided during worksheet completion.

Resources and Materials & Technology Integration -- List ALL items necessary for the delivery of the lesson. Include any attachments of learner worksheets used and teacher support material that will support communication of instruction.

Day 1 – Ribbon Measurement Activity
Ribbons or string (one per pair, approximately 50–60 cm each)

Scissors

Rulers (cm)

6–8 circular objects of varying sizes per group (lids, cups, rolls of tape, jar tops, paper plates, cans, small containers)

Recording sheet (table with columns: Object Name, Circumference, Diameter, Circumference ÷ Diameter)

Calculators (one per pair)

Chart paper or whiteboard for class data collection

Day 2 – Circle Terminology
Circle Terminology Worksheet (diagram labelling + matching definitions)

Whiteboard and markers

Printed or projected diagram of a circle with parts labelled (anchor chart)

Student notebooks

Compasses (optional, for drawing circles)

Day 3 – Calculating Diameter, Radius, and Circumference
Circumference and Radius/Diameter Calculations Worksheet (practice problems)

Whiteboard and markers

Calculators

Anchor chart from Day 2 (posted in classroom)

Day 4 – Deriving the Area of a Circle
Pre-cut paper circles (one per student or pair, approximately 15–20 cm diameter)

Scissors (one per student)

Glue sticks

Blank paper for gluing sectors

Area of a Circle Worksheet (derivation explanation + calculation problems)

Whiteboard and markers

Calculators

Three Part Lesson

Indicate Instructional Strategy -- modelled, shared, guided, independent instruction, small group, conferencing, whole class, other _____

Day 1: Ribbon measurement activity where students discover π by measuring circumference and diameter of circular objects, recording data, and finding the constant ratio

Day 2: Explicit teaching of formal terminology (radius, diameter, circumference, centre, chord, pi) with an anchor chart, guided practice, and a Circle Terminology Worksheet

Day 3: Modelled examples and guided practice calculating circumference, diameter, and radius using $C = \pi d$ and $C = 2\pi r$, with a Circumference Calculations Worksheet

Day 4: Problem-based learning task where students cut paper circles into sectors, rearrange them into a rectangle, and derive $A = \pi r^2$ from background knowledge, followed by an Area of a Circle Worksheet

Going with the flow but above highlight are the plans for each day. Going to be using the worksheets to help guide the students during this lesson.

Lessons are derived from the following website: https://learn.makemathmoments.com/task/going-in-circles/

Day 1: Ribbon Measurement Activity – Discovering Pi (50 minutes)

Lesson Title: Discovering the Relationship Between Circumference and Diameter
Instructional Strategies: Guided inquiry, small group, whole class discussion

Minds-On (10 minutes)
Teacher does: Hold up a large circular object (e.g., a paper plate or lid). Ask the class:

"If I asked you to describe the size of this circle to someone who cannot see it, what would you measure? How would you measure it?"

Take 3–4 quick responses. Students may say things like "measure across it," "measure around it," or "measure from the middle." Do not correct or introduce vocabulary yet — simply acknowledge their ideas.

Teacher says: "Today, you are going to be detectives. I am going to give each pair a ribbon, a ruler, and some circular objects. Your job is to measure the distance around each circle and the distance across each circle, record your data, and see if you can discover a hidden pattern."

Prompting Questions:

What measurement would you choose to describe the size of this circle, and why?

Is there more than one way to describe how big a circle is?

Action (30 minutes)
Set-Up (5 minutes):

Teacher does: Organize students into pairs. Distribute to each pair: one ribbon/string, one ruler, one calculator, one recording sheet, and 3–4 circular objects.

Teacher says: "Here is what you are going to do. First, take your ribbon and wrap it carefully around the outside edge of your circular object — that is the distance all the way around. Mark or cut the ribbon where it meets, then lay it flat against your ruler to measure it in centimetres. Record this in the 'Circumference' column on your recording sheet."

Teacher says: "Next, take your ribbon and stretch it across the widest part of the circle — right through the middle. Measure that with your ruler. Record it in the 'Diameter' column."

Teacher says: "Finally, use your calculator to divide the circumference by the diameter. Record your answer to two decimal places in the last column."

Model this explicitly: Demonstrate the full process with one circular object in front of the class. Wrap the ribbon around, cut/mark it, measure it, record it. Then measure across, record it. Then divide on the calculator. Write the example on the board.

Student Work (20 minutes):

Students work in pairs to measure 3–4 circular objects each. Teacher circulates actively and asks:

"Is your ribbon tight around the object, or is there slack? Why does that matter?"

"How did you find the widest part across? How do you know it goes through the middle?"

"What number are you getting when you divide? Is it similar to your other objects?"

"Compare your ratio with the pair beside you. What do you notice?"

Whole-Class Data Collection (5 minutes):

Teacher does: Call students back. Ask each pair to share one of their circumference ÷ diameter ratios. Record all ratios on the board or chart paper in a visible list.

Teacher says: "Look at all these numbers on the board. What do you notice?"

Students should observe that the ratios are all close to 3.1 or 3.14. Guide them to see that this is always roughly the same number, regardless of circle size.

Teacher says: "This number that keeps showing up — about 3.14 — is one of the most famous numbers in all of mathematics. It is called pi, and we write it with the Greek letter π. Pi is the ratio of the circumference to the diameter for every circle in the world."

Consolidation (10 minutes)
Teacher does: Write on the board: Circumference ÷ Diameter = $\pi \approx 3.14$

Teacher says: "So if we know the diameter of any circle, how could we figure out the circumference?" (Guide students to see: Circumference = π × Diameter, or $C = \pi d$.)

Teacher says: "And if the diameter is twice the radius, how else could we write this?" (Guide to: $C = 2\pi r$.)

Ask students to write in their notebooks:

What pattern did you discover today?

Why do you think this ratio is always the same, no matter the size of the circle?

Prompting Questions for Consolidation:

What did every group's ratio have in common?

Why does it make sense that a bigger circle has a bigger circumference but the same ratio?

If I told you a circle has a diameter of 10 cm, could you estimate the circumference right now?

Extension Activity: Students who finish early can measure additional objects, or try to predict the circumference of a new object by measuring only the diameter and multiplying by π, then verify by wrapping the ribbon.

Next Steps: Tomorrow we will learn the formal mathematical names for all the parts of a circle.

Day 2: Circle Terminology – Radius, Diameter, Circumference, and More (50 minutes)
Lesson Title: Circle Vocabulary and Terminology
Instructional Strategies: Modelled, shared, guided, independent instruction, whole class

Minds-On (8 minutes)
Teacher does: Draw a large circle on the board. Do not label anything yet.

Teacher says: "Yesterday we measured circles using ribbons. We measured the distance across and the distance around. Today we are going to learn the proper mathematical names for all the parts of a circle. Before I teach you, let's see what you already know."

Think-Pair-Share: Ask students to turn to a partner and list as many parts of a circle as they can think of (from yesterday's activity or prior knowledge). Give 2 minutes, then take responses. Record student-generated terms on the board (e.g., "across," "around," "middle").

Teacher says: "Great thinking. Now let's match your everyday language to the official math vocabulary."

Action (30 minutes)
Explicit Teaching – I Do (15 minutes):

Teacher does: Draw a clean, large circle on the board. Introduce each term one at a time, drawing and labelling as you go:

Centre: "The centre is the exact middle point of the circle. Every point on the circle is the same distance from the centre. I am going to mark it with a dot and label it 'centre.'"

Radius: "The radius is the distance from the centre to any point on the edge of the circle." Draw a line segment from the centre to the edge. Label it. "Yesterday, this would have been half of the distance across. The plural of radius is radii."

Diameter: "The diameter is the distance across the circle, passing through the centre." Draw a line segment all the way across through the centre. Label it. "Notice the diameter is exactly two radii. So diameter = 2 × radius, and radius = diameter ÷ 2."

Circumference: "The circumference is the distance all the way around the circle — the perimeter of a circle. This is what you measured with your ribbon yesterday."

Chord: "A chord is any straight line segment that connects two points on the circle. The diameter is the longest chord." Draw a chord that does not pass through the centre.

Pi (π): "Pi is the ratio of the circumference to the diameter. It is approximately 3.14. It is the same for every circle. We write it as the Greek letter π."

Teacher does: Create an anchor chart (or reveal a pre-made one) with a diagram showing all labelled parts. Post it in the classroom for the rest of the unit.

We Do – Guided Practice (5 minutes):

Teacher does: Draw 2–3 new circles on the board. For each one, point to a part and ask the class to name it. Then give a measurement (e.g., "The radius is 4 cm. What is the diameter?") and have students respond.

Practice examples:

Radius = 5 cm → Diameter = ? (10 cm)

Diameter = 18 cm → Radius = ? (9 cm)

Point to the edge and ask: "What do we call the distance all the way around?" (Circumference)

You Do – Independent Practice with Worksheet (10 minutes):

Teacher says: "Now you are going to complete the Circle Terminology Worksheet. On this worksheet you will label diagrams of circles with the correct terms, match definitions to vocabulary words, and answer a few short-answer questions about the relationships between radius, diameter, and circumference."

Circle Terminology Worksheet includes:

A blank circle diagram where students must label: centre, radius, diameter, circumference, chord

A matching section: match the term to its definition (6 terms, 6 definitions)

3 short-answer questions:

"If the radius of a circle is 7 cm, what is the diameter? Explain how you know."

"What is the relationship between the circumference and the diameter?"

"Is every diameter also a chord? Is every chord a diameter? Explain."

Teacher does: Circulate during independent work. Check for understanding. Address common misconceptions (e.g., students confusing radius and diameter, or thinking a chord must pass through the centre).

Consolidation (12 minutes)
Teacher does: Take up key worksheet questions as a class. Call on students to share their labelled diagrams and answers.

Exit Ticket (last 5 minutes): On a sticky note, students answer:

Define "radius" in your own words.

A circle has a diameter of 20 cm. What is the radius?

Name one thing you learned today that you did not know before.

Collect sticky notes as students leave.

Prompting Questions:

How are the radius and diameter related? Can you describe this in your own words?

If you know the radius, what else can you figure out?

Why is the diameter the longest chord?

Extension Activity: Students who finish the worksheet early can draw their own circles using a compass and label all parts, or write a "Circle Vocabulary Story" that uses all six terms in context.

Next Steps: Tomorrow we will use these terms and the formulas to calculate circumference, diameter, and radius.

Day 3: Calculating Diameter, Radius, and Circumference (50 minutes)
Lesson Title: Applying Circle Formulas – Circumference, Diameter, and Radius
Instructional Strategies: Modelled, guided, independent instruction, whole class

Minds-On (7 minutes)
Teacher does: Display the anchor chart from Day 2. Quick review:

Teacher says: "Before we start calculating today, let's do a quick warm-up. On your mini whiteboard (or in your notebook), answer these three questions:"

The radius of a circle is 6 cm. What is the diameter?

The diameter of a circle is 15 cm. What is the radius?

What is the approximate value of π?

Take answers quickly. Address any errors. Confirm: diameter = 2r; radius = d ÷ 2; π ≈ 3.14.

Teacher says: "Today we are going to use these relationships and the value of pi to calculate the circumference of circles — and to work backwards from the circumference to find the diameter or radius."

Action (33 minutes)
Explicit Teaching – I Do (15 minutes):

Teacher does: Write the two circumference formulas on the board:

$C = \pi d$ (Circumference = pi × diameter)
$C = 2\pi r$ (Circumference = 2 × pi × radius)

Worked Example 1 — Finding Circumference Given Diameter:

"A circle has a diameter of 12 cm. Find the circumference."

$C = \pi d$

$C = 3.14 \times 12$

$C = 37.68$ cm

Teacher says: "I substituted the diameter into the formula and multiplied by pi. Always show your formula, your substitution, and your answer with units."

Worked Example 2 — Finding Circumference Given Radius:

"A circle has a radius of 5 cm. Find the circumference."

$C = 2\pi r$

$C = 2 \times 3.14 \times 5$

$C = 31.4$ cm

Teacher says: "Since I was given the radius, I used $C = 2\pi r$. I could also have found the diameter first (10 cm) and used $C = \pi d$. Both give the same answer."

Worked Example 3 — Finding Diameter Given Circumference:

"A circle has a circumference of 47.1 cm. Find the diameter."

$C = \pi d$, so $d = C \div \pi$

$d = 47.1 \div 3.14$

$d = 15$ cm

Teacher says: "To find the diameter, I rearranged the formula. If $C = \pi d$, then $d = C \div \pi$. Always think about what you are solving for."

Worked Example 4 — Finding Radius Given Circumference:

"A circle has a circumference of 25.12 cm. Find the radius."

$d = C \div \pi = 25.12 \div 3.14 = 8$ cm

$r = d \div 2 = 4$ cm

We Do – Guided Practice (8 minutes):

Teacher does: Put 3 problems on the board. Solve them together as a class, calling on students to provide each step:

Diameter = 20 cm. Find C. (62.8 cm)

Radius = 9 cm. Find C. (56.52 cm)

Circumference = 94.2 cm. Find d and r. (d = 30 cm, r = 15 cm)

For each, ask: "What formula do I use? What do I substitute? What is my answer with units?"

You Do – Independent Worksheet (10 minutes):

Teacher says: "Now you will complete the Circumference and Radius/Diameter Calculations Worksheet independently. Show your work for every question: write the formula, substitute, solve, and include units."

Worksheet includes (approximately 10–12 questions):

3 questions: Given diameter, find circumference

3 questions: Given radius, find circumference

2 questions: Given circumference, find diameter

2 questions: Given circumference, find radius

2 word problems (e.g., "A circular garden has a radius of 3.5 m. How much fencing is needed to go around it?" and "A bike wheel has a circumference of 188.4 cm. What is the diameter of the wheel?")

Teacher does: Circulate during independent work. Provide individual feedback. Look for common errors: forgetting to multiply by 2 when using $C = 2\pi r$, confusing radius and diameter, or forgetting units.

Consolidation (10 minutes)
Teacher does: Take up 3–4 key worksheet problems on the board, choosing questions that students found challenging. Have students volunteer to explain their solutions.

Key Takeaways — Write on the board:

$C = \pi d$ or $C = 2\pi r$

$d = C \div \pi$

$r = d \div 2$

Teacher says: "Tomorrow, we are going to figure out how to find the area inside a circle. We already know how to find the area of a rectangle. Tomorrow, we will use that knowledge to unlock the area of a circle."

Prompting Questions:

When do you use $C = \pi d$ versus $C = 2\pi r$? Does it matter?

If the circumference doubles, what happens to the diameter? Why?

How could you check your answer?

Extension Activity: Students who finish early can solve challenge problems (e.g., "Two circles have radii of 4 cm and 8 cm. How much greater is the circumference of the larger circle?") or create their own word problem for a partner.

Next Steps: Tomorrow we will derive the area formula by cutting circles into sectors.

Day 4: Deriving the Area of a Circle – Problem-Based Learning (50 minutes)
Lesson Title: Discovering the Area of a Circle by Cutting Sectors
Instructional Strategies: Problem-based learning, modelled, guided, small group, whole class

Minds-On (8 minutes)
Teacher does: Quick review. Write on the board:

Area of a Rectangle = length × width

Teacher says: "We already know how to find the area of a rectangle. Today, I am going to give you a challenge. I am going to give each pair a paper circle. Your challenge is to figure out a way to find the area of that circle — but here is the catch: you are only allowed to use what you already know. You know the area of a rectangle. Can you somehow turn a circle into a rectangle?"

Pause and let students think for 30 seconds. Take 2–3 predictions.

Teacher says: "Let's see if we can figure this out together using scissors and our background knowledge."

Prompting Questions:

What shapes do you already know how to find the area of?

Could you cut a circle into pieces and rearrange them into a shape you recognize?

Action (32 minutes)
Problem-Based Inquiry – Cutting Sectors (15 minutes):

Teacher does: Distribute one pre-cut paper circle, scissors, glue stick, and a blank piece of paper to each pair.

Step 1 — Teacher models: "Watch me first. I am going to cut this circle into equal sectors — like slices of a pizza." Cut the circle into 8 equal sectors (or 12 for better approximation). Show the class your 8 "pizza slices."

Step 2 — Teacher models rearranging: "Now I am going to take these sectors and arrange them by alternating them — one pointing up, one pointing down — and slide them together." Arrange the sectors into an approximate parallelogram/rectangle shape on the blank paper. "What shape does this look like?"

Students should say "rectangle" or "parallelogram."

Step 3 — Teacher says: "Now it is your turn. Cut your circle into at least 8 equal sectors. Then arrange them into a rectangle shape and glue them down on your blank paper."

Students work in pairs to cut and rearrange. Teacher circulates.

Explicit Teaching – Deriving the Formula (10 minutes):

Once students have their "rectangles" glued down, bring the class back together.

Teacher does: Draw the rearranged shape on the board — a bumpy-edged rectangle.

Teacher says: "Let's figure out the dimensions of this rectangle. Look at the height first. Each sector is shaped like a thin triangle. The height of each sector is the distance from the edge of the circle to the centre. What is that called?" (Radius.)

Teacher writes: Height = r

Teacher says: "Now look at the width — the long bumpy edge. That bumpy edge is made up of the curved parts from the outside of the circle. The top bumps are from half the circumference, and the bottom bumps are from the other half. So the width is approximately half the circumference."

Teacher writes: Width = ½ × Circumference = ½ × $2\pi r$ = πr

Teacher says: "So if the area of a rectangle is length times width, then:"

Teacher writes on board:

Area = height × width

Area = $r \times \pi r$

Area = πr^2

Teacher says: "The more sectors we cut, the closer our shape gets to a perfect rectangle, and the more exact the formula becomes. This is how mathematicians derived the formula for the area of a circle — using the area of a rectangle."

Guided Practice – We Do (3 minutes):

Solve one example together:

"A circle has a radius of 7 cm. Find the area."

$A = \pi r^2$

$A = 3.14 \times 7^2$

$A = 3.14 \times 49$

$A = 153.86\ cm^2$

Teacher says: "Remember: the radius gets squared first, then multiplied by pi. And area is always in square units."

Independent Practice — Area Worksheet (4 minutes to start, finish for homework if needed):

Teacher says: "Complete the Area of a Circle Worksheet. For each question, write the formula, substitute, solve, and include units."

Worksheet includes:

2 questions: Given radius, find area

2 questions: Given diameter, find area (students must first find the radius)

1 question: Given area, find radius (working backwards: $r = \sqrt{(A \div \pi)}$)

2 word problems (e.g., "A circular swimming pool has a diameter of 8 m. What is the area of the pool cover needed?" and "A pizza has a radius of 15 cm. What is the area of the top of the pizza?")

1 explanation question: "In your own words, explain how cutting a circle into sectors and rearranging them into a rectangle helps us find the formula $A = \pi r^2$. Use a diagram."

Consolidation (10 minutes)
Teacher does: Take up 2 key worksheet problems. Then return to the big picture:

Teacher says: "Let's look back at our whole unit. On Day 1, we used ribbons to discover that the circumference divided by the diameter always gives us about 3.14 — that is pi. On Day 2, we learned the proper names: radius, diameter, circumference, chord, and centre. On Day 3, we calculated circumference using $C = \pi d$ and $C = 2\pi r$. And today, we used what we know about rectangles to discover that the area of a circle is $A = \pi r^2$."

Write the formula summary on the board:

$d = 2r$

$C = \pi d = 2\pi r$

$A = \pi r^2$

Exit Ticket: On a sticky note, students answer:

What is the area of a circle with a radius of 10 cm?

In one sentence, explain how a circle can be turned into a rectangle to find area.

Collect as students leave.

Prompting Questions:

Why does cutting the circle into more sectors make the rectangle more accurate?

What would happen if you cut the circle into 100 sectors? 1,000?

How are circumference and area different? Do they use the same measurements?

Extension Activities -- What will learners do when work is completed? What will learners do if they finish early?

Students who finish early can explore: "If the radius of a circle doubles, does the area double? Prove it with an example." (Answer: No — the area quadruples because area depends on r^2.)

Next Steps: Future lessons will extend to constructing circles with a compass given specific measurements (E2.4), and calculating the surface area of cylinders (E2.6).

Next Steps -- Where will this lesson lead to next?

My Lesson Reflection - Placement Only

This portion of the lesson will be completed following the conclusion of the lesson (what went well, what will I change - see reflection questions)